

Interpretable Lightweight CNN for Colon and Lung Cancer Classification with LIME based Explainability

Nishchal Adil
Computer Science and Engineering
National Institute of Technology
Raipur, India
nadil.phd2023.cs@nitrr.ac.in

Pradeep Singh
Computer Science and Engineering
National Institute of Technology
Raipur, India
psingh.cs@nitrr.ac.in

Naresh Kumar Nagwani
Computer Science and Engineering
National Institute of Technology
Raipur, India
nknagwani.cs@nitrr.ac.in

Abstract— Colon and lung cancers are one of the major reasons reported for cancer related death across the globe, which underscores the vital need for early and accurate method for its diagnosis. Our study presents a lightweight CNN model for classification of colon and lung cancer disease using histopathological images. The proposed model fuses Convolutional Block Attention Module (CBAM) within adaptive residual blocks where skip connections are created by adding input tensor to the output of residual block, so as to enhance feature extraction and refine the feature map through inclusion of attention mechanism. Comprehensive evaluations on LC25000 dataset for both lung and colon cancers demonstrate the model's effectiveness in classification, with overall accuracy of 99.8%. In spite of its compact architecture with only 0.9M trainable parameters our model achieved high classification accuracy. Further to facilitate transparency and interpretability Local Interpretable Model-agnostic Explanation (LIME) is being generated which provides insights into the region which contributes to model's decision. The incorporation of LIME facilitates a deeper comprehension of the model's behavior, supporting its potential for clinical adoption and improving diagnostic reliability.

Keywords—: Lung and colon Cancer, CBAM, Convolutional Neural Networks, LIME, Histopathological Images

I. INTRODUCTION

Healthcare is seen as a crucial aspect in sustaining the health and quality of life for individuals and communities globally. Access to good healthcare has been regarded as an essential right that is vital to promote good health and prevent from several diseases. Reliable and trustworthy healthcare infrastructure is of the utmost importance to tackle diverse public health conditions and meeting the needs of vulnerable populations. Recent advancements in technology and medicine significantly improves healthcare systems, making them more affordable and improving healthcare administration. This remarkable growth has led to advances in new therapies and medicines for serious medical conditions such as cancer, which holds the second highest spot in worldwide mortality statistics [1].

Colon and lung cancers is ranked among the most prevalent and potentially dangerous malignancies worldwide, severely impacting death rates due to cancer. Lung cancer constitutes the predominant reason for death related to cancer, with an approximated 2.2 million newly diagnosed cases and 1.8 million fatalities every year. The

elevated mortality rate is primarily due to delayed diagnosis, as initial symptoms can be insignificant or misidentified as other respiratory ailments. Colorectal cancer, impacts the lower portion of the digestive system, ranked as the third most prevalent cancer worldwide and is the second foremost cause of death from cancer. Timely identification of tumors in the colon may substantially improve survival rates; however, similar to lung cancer, it frequently remains undiagnosed until the disease becomes severe [2].

Early and accurate identification of both lung and colon cancer is essential, but using traditional methods, like medical imaging, colonoscopy, and biopsy, are time intensive task, invasive in nature, and sometimes fail to diagnose the disease at initial stage. This has sparked the interest of researchers to deploy deep learning models, particularly CNNs, to facilitate automated classification of cancer from medical images. CNNs can detect unidentifiable patterns and abnormalities in lung and colon tissues, potentially enabling earlier and more precise diagnoses than traditional methods alone [3].

Though the use of CNN for medical diagnosis and applications has become very popular, it is black box in nature which limits its acceptance and adaptation in clinical setup. Taking this into consideration explainable AI has emerged as one of the most promising areas of research in healthcare domain. Advanced XAI methods facilitate deep learning models to be more transparent and interpretable, providing insights into how and why a particular prediction is made, it is particularly very important when dealing with some perilous disease such as lung and colon cancer [4], [5]. This escalated the use of XAI techniques like LIME, Grad-CAM, SHAP to fill this gap by generating visualizations model's decisions and making deep learning models more transparent and trustworthy in clinical settings [6].

This study is aimed to create a lightweight CNN model for the classification of lung and colon cancer disease, and to generate visual explanations LIME is being incorporated. Our study is not only limited to improve diagnostics efficiency of a deep learning model but also build up trust on AI based healthcare systems and assure that clinicians can interpret and rely on the model's outputs.

II. RELATED WORK

Wani et al. [7] introduced DeepXplainer, which is an explainable CNN-based model for lung cancer detection that has a structure consisting of CNN as the model to learn

features, XGBoost as the learning classifier and SHAP (Shapley Additive explanations) for generating interpretable explanations. Venkata Krishnamoorthy et al. [8] aimed to improve the diagnosis of diseases such as COVID-19, Pneumonia, and Tuberculosis. The methodology integrates pre-processing with DFHE, feature extraction through an Adaptive Attention-Based Deep Neural Network and classification using DenseConvNASNet while it provides a comprehensive explanation from LIME. Islam et al. [9] proposed an DCNN-ViT-GRU model for lung cancer, COVID-19 and pneumonia detection and classification and incorporate explainable AI technique to facilitate interpretability. They proposed a hybrid DCNN-ViT-GRU model for feature extraction and classification and used LIME and SHAP as XAI techniques to improve the interpretability and transparency of the model's decision-making process. Moin et al. [10] used pre-trained CNN models like Xception, DenseNet (121, 169, 201), ResNet (101, 152), InceptionV3, and InceptionResNetV2 to improve classification accuracy through enhanced data augmentation techniques. Additionally, they employed attention visualization methods such as GradCAM, GradCAM++, ScoreCAM, LayerCAM, and SmoothGrad to enhance interpretability, providing valuable insights into the models' classification of malignant and benign images. Tummala et al. [11] proposed a fully automated framework using EfficientNetV2 models to classify 5 lung and colon cancer subtypes from histopathology images, also obtained visual saliency maps generated using GradCAM to help pathologists understand the model's decision-making process. Civit-Masot et al. [12] designed and implemented a deep learning-based medical diagnostic system for non-small cell lung cancer identification using histopathological images and includes an explainable deep learning module to provide information to the pathologist.

III. PROPOSED METHODOLOGY

In this section we introduce our proposed work, initially the Channel Attention (CA) mechanism is discussed, which enhances the importance of feature maps by applying global average and max pooling followed by dense layers and a sigmoid activation. Next, it describes the Spatial Attention (SA) mechanism, which identifies important regions in the spatial dimensions by applying pooling operations, concatenation, and a convolutional layer. Then, CBAM (Convolutional Block Attention Module) block combines both CA and SA to refine feature representations. As discussed in later section the CBAM module is integrated into each adaptive residual block of the model, which also includes skip connections to preserve information. Finally, the full model architecture which infuses CBAM into adaptive residual blocks (ARB), perform down-sampling, and dense layers, with softmax activation for obtaining class predictions.

A. Channel Attention

The channel attention (CA) mechanism is designed to focus the most important feature maps along the channel dimension. This mechanism performs global average pooling (GAP) and max pooling operations across the

spatial dimensions, compressing each feature map into a single value. The values are then passed through a set of Dense layers (FC layers) to obtain the channel attention weights. These weights are then applied to the input tensor by obtaining its Hadamard product (element-wise multiplication).

The equation for channel attention is:

$$F_{ca} = \sigma \left(\begin{array}{l} \text{Dense} \left(\text{ReLU} \left(\text{Dense}(\text{AvgPool}(F)) \right) \right) \\ + \text{Dense} \left(\text{ReLU} \left(\text{Dense}(\text{MaxPool}(F)) \right) \right) \end{array} \right) \quad (1)$$

where:

- F_{ca} denotes the channel attention (CA) output,
- $\text{AvgPool}(F)$ and $\text{MaxPool}(F)$ are global average and max pooling across the spatial dimensions of feature tensor F ,
- Dense layers reduce and then restore the dimensionality of the feature maps,
- σ denotes the sigmoid activation to scale the attention weights between 0 and 1.

B. Spatial Attention

After applying channel attention, the spatial attention mechanism identifies which regions in the spatial dimensions (height and width) are most important. This module pools the input feature maps along the channel dimension using average pooling and max pooling operations, resulting in two spatial attention maps. The resultant maps are concatenated and made to pass through a 2D convolutional layer with a sigmoid nonlinearity function to produce a spatial attention map. The spatial attention output is computed as:

$$F_{sa} = \sigma(\text{Conv2D}([\text{AvgPool}(F), \text{MaxPool}(F)])) \quad (2)$$

where:

- F_{sa} gives the spatial attention (SA) output,
- $\text{AvgPool}(F)$ and $\text{MaxPool}(F)$ reduce the input feature maps along the channel axis,
- The concatenated result is processed through a 7x7 convolution to aggregate spatial attention,
- σ is the sigmoid non-linearity applied to scale the spatial attention.

C. CBAM block

The CBAM block combines channel attention and spatial attention to sequentially refine the feature maps. First, the input tensor passes through the channel attention block, followed by the spatial attention block. This combination allows the model to highlight both the most informative channels and the most important spatial locations [13]. The output of the CBAM block is:

$$F_{cbam} = F_{sa}(F_{ca}(F)) \quad (3)$$

where:

- $F_{ca}(F)$ is the feature tensor after channel attention,
- $F_{sa}(\cdot)$ applies spatial attention to the result of channel attention.

D. Adaptive Residual Block

In this architecture, each residual block incorporates the CBAM block to enhance the feature representation. The residual block performs two convolutions followed by batch normalization and applies the CBAM block to the output. A skip connection is added by projecting the input tensor into the same dimension as the output of the convolutions using a 1x1 convolution. The final output of the residual block is obtained by adding the output of the CBAM block to the skip connection:

$$F_{residual} = F_{cbam} + F_{skip} \quad (4)$$

where:

- F_{cbam} is the output of the CBAM block,
- F_{skip} is the shortcut connection (projected input tensor).

E. Model Architecture

The model begins with an initial convolutional layer, followed by multiple residual blocks, each integrated with CBAM blocks to emphasize important features. The model progressively down samples the spatial resolution through max pooling after each residual block. After the feature extraction stages, the model flattens the output and passes it through fully connected layers, followed by a dropout layer to mitigate overfitting. Finally, the model outputs the predicted class probabilities using a softmax activation function. Detailed architecture is shown in Figure 1. Mathematically, the output of the model is:

$$\hat{y} = \text{softmax} \left(\left(M_2 \cdot \text{ReLU} \left(M_1 \cdot \text{Flatten}(F_{final}) \right) + \theta_1 \right) + \theta_2 \right) \quad (5)$$

where:

- F_{final} is the feature tensor after all the CBAM-integrated residual blocks,
- M_1 and M_2 are weight matrices of the FC layers,
- θ_1 and θ_2 represents bias terms.

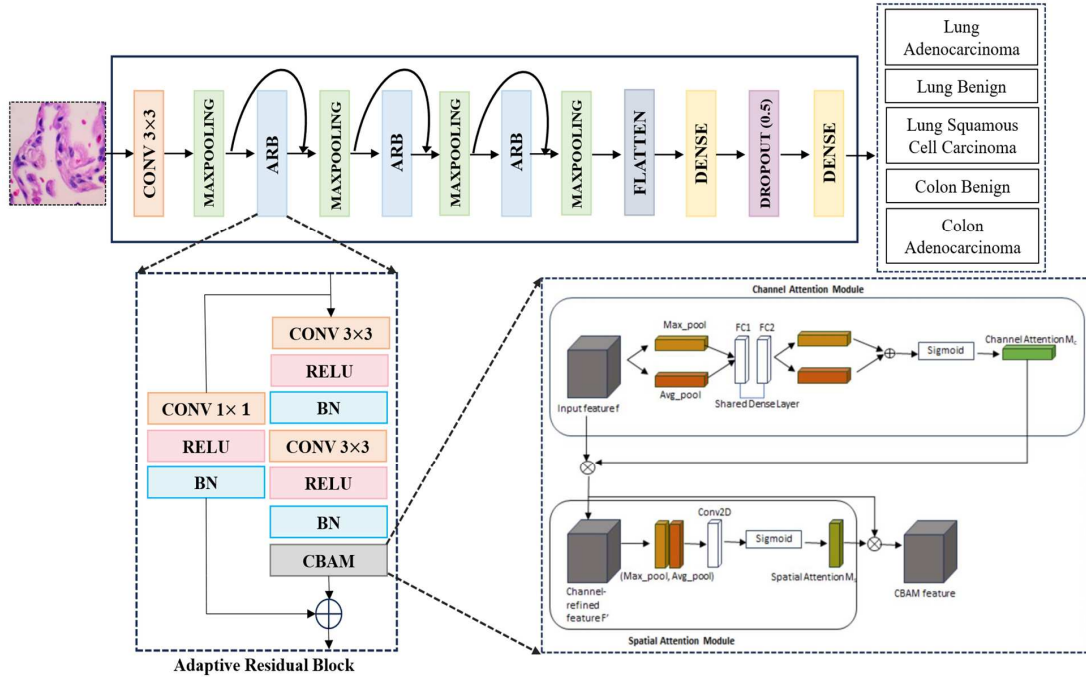


Fig. 1. Proposed model architecture

IV. RESULT AND DISCUSSION

A. Dataset Description

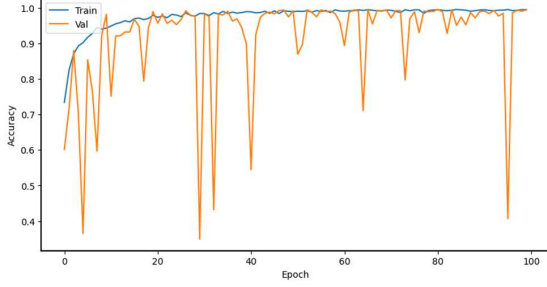
In this study, the LC25000 dataset is used [14], which is constituted of 25,000 images of colon and lung tissues, classified into 5 categories. The colon images are classified into two categories: adenocarcinoma (colon_aca), and benign tissues (colon_n), while the lung tissue histopathological images are divided into 3 categories: adenocarcinoma (lung_aca), squamous cell carcinoma (lung_scc), and benign (lung_n). All 5 classes in the dataset

consist of 5000 images of each category. So, we are using all 25000 images, as mentioned earlier we are dealing with multi-class classification, in our work we are using 70% data for training (17500 samples), 15% data for validation (3750 samples), and 15% data for testing purpose (3750 samples). All images were resized to 150×150 dimension to make it suitable for classification task.

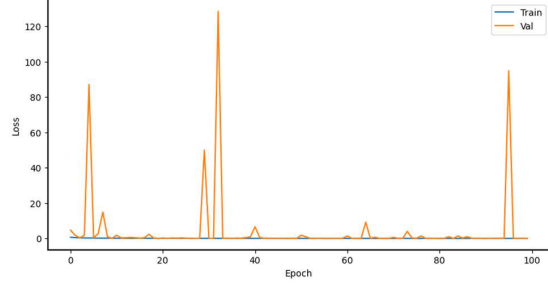
B. Experimental results

A custom convolutional neural network model was created with an input shape of $150 \times 150 \times 3$ and five output

classes. The model was compiled using the Adam optimizer, as our model is performing multi class classification categorical cross-entropy loss function is used, and accuracy is used as the evaluation metric. Training was performed over 100 epochs with a batch size of 16. The classification result shows the model's performance on a test dataset, providing key metrics such as precision, recall, and F1-score for each class. The model exhibits high precision, recall, and F1-scores across all five classes, including benign, malignant adenocarcinoma, malignant squamous cell carcinoma, benign colon tissue, and colon adenocarcinoma. With precision, recall, and F1-scores consistently near 1.00,



(a)



(b)

Fig. 2. (a)Model Accuracy (b) Model Loss

C. Comparison analysis on trained models

This section shows the comparison analysis of proposed method with other trained attention mechanisms. In this study to analyze the effect of various attention mechanisms we have replaced Convolutional Block Attention Model [13] with Squeeze and Excitation block [15] and Global Context Network block [16]. Table I shows class wise performance of trained models.

TABLE I. CLASSWISE PERFORMANCE OF TRAINED MODELS

Measure	lung_n	mal_aca	mal_scc	colon_n	colon_aca
Model 1 (CNN+CBAM)					
Precision	1	0.99	0.99	1	1
Sensitivity (Recall)	1	0.99	0.99	1	1
F1-Measure	1	0.99	0.99	1	1
Model 2 (CNN+SE)					
Precision	1	1	0.96	1	1
Sensitivity (Recall)	1	0.96	1	1	1
F1-Measure	1	0.98	0.98	1	1
Model 3 (CNN+GCNet)					
Precision	1	1	0.88	1	1
Sensitivity (Recall)	1	0.86	1	1	1
F1-Measure	1	0.92	0.93	1	1

For the lung_n and colon_aca classes, all three models CBAM, SE, and Global Context Network demonstrate perfect performance, achieving 100% in precision, recall, and F1-score. This indicates that the models can accurately identify these classes without any misclassifications. For class malignant adenocarcinoma (lung_aca), results shows

the model demonstrates almost no false positives or false negatives. The accuracy of 99.8% indicates that the model correctly classified 99.8% of the 3,750 test samples. The macro average and weighted average of precision, recall, and F1-scores, both around 0.99, further confirm that the model performs uniformly well across all classes, even those with fewer instances. These results highlight the robustness and reliability of the model in distinguishing between different tissue types, making it highly effective for the classification task. Figure 2 shows the plot of model's accuracy and loss during training and validation.

that CBAM demonstrate high precision and and sensitivity(recall) of 99%, whereas SE block exhibits comparatively lower sensitivity of 96%. The GCNet block showed lowest performance for this category with sensitivity of 86% and F1-Measure of 92%. By this we can infer that GCNet struggles to correctly identify all instances resulting in higher number of false negative instances. For Malignant Squamous Cell Carcinoma (lung_scc). CBAM and SE attained good performance, CBAM obtained precision and recall of 99%, whereas SE showed a slight drop in precision 96% which is compensated with recall of 100%. However Global Context Network underperforms in this class, with only precision of 88% and recall of 93%. Through this, it can be stated that Global Context Network is misclassifying instances of this category. Finally for colon benign(colon_n) all 3 models achieved 100% precision, recall, and F1-Measure depicting its reliability to identify colon benign tissue. This demonstrates the robustness of the models in distinguishing benign tissue from cancerous types.

TABLE II. OVERALL PERFORMANCE OF TRAINED MODELS

Metric	CNN+CBAM	CNN+SE	CNN+GCNet
Test Accuracy	0.998	0.992	0.971
Precision	0.99	0.99	0.97
Sensitivity (Recall)	0.99	0.99	0.97
F1-Measure	0.99	0.99	0.97

Table II compares the overall performance of the CBAM, SE, and Global Context Network models. CBAM achieved the highest accuracy at 99.8%, followed by SE at 99.2%, and Global Context Network at 97.1%. CBAM and SE also scored identical macro precision, recall, and F1-scores (0.99), indicating consistent performance across all classes. In contrast, the Global Context Network scored slightly lower at 0.97 for these metrics, reflecting more variability in its class-specific predictions. Overall, CBAM and SE outperform the Global Context Network in classification accuracy and consistency.

The confusion matrix shown in Fig.3 depicts comparison and highlights the strengths and weaknesses of the CBAM, SE, and Global Context Network models

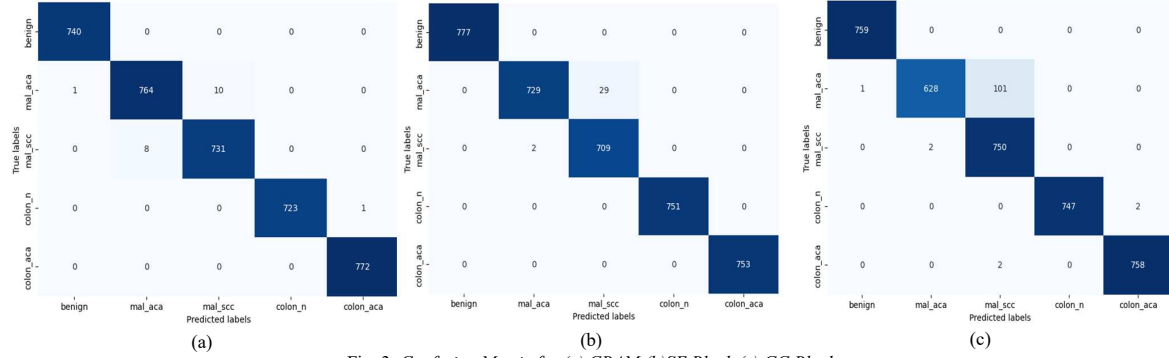


Fig. 3. Confusion Matrix for (a) CBAM (b)SE Block (c) GC Block

D. LIME Visualization

In order to generate visual explanations from model's predictions on histopathological images we used Local Interpretable Model-agnostic Explanations framework. LIME produces superpixels by perturbing the provided input image and analyze model's output and thereby representing most important region which effects the model's classification result. we resized and normalized input image before feeding it to custom prediction function to interact with model. Here LIME is configured to provide an explanation by 1000 samples of perturbed images and identify most important superpixels contributing to prediction. Result of LIME highlights these superpixels in color providing visual of most influential region for making prediction. Additionally, we illustrated the superpixel importance values through horizontal bar graph. Positive scores are represented using green color implying regions contributing positively to the model's classification and negative scores are depicted using red color showing regions that detracted from the classification. The differentiation between positive (green) and negative (red) contributions, as illustrated in Figure 4, provides more deeper understanding. This thorough interpretability analysis provides transparency in the model's decision-making process, which is significantly important for the relying of medical image classification tasks.

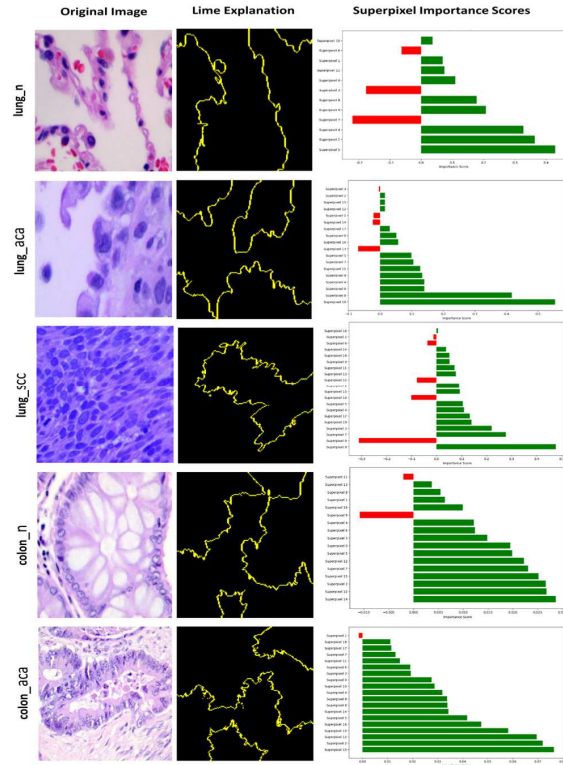


Fig. 4. Results of Lime visualizations and its plot of superpixel importance score

V. CONCLUSION

In this study, we propose lightweight CNN model incorporated with CBAM and skip connections with in adaptive residual block for lung and colon cancer classification task, our model achieved outstanding result with an overall accuracy of 99.8%. Inclusion of attention mechanism like CBAM significantly improves model's ability to focus only on specific region, thereby enhancing its precision and recall across multiple classes of cancer types. One of the main advantages of proposed model is ability to maintain high accuracy while maintaining low computational requirement, making it suitable for deployment in resource constraint medical settings. Additionally, LIME explanations are added to enhance interpretability and transparency to understand model's decision making. However, addition of CBAM block may increase its complexity leading to longer inference time. In future research this work can be extended by applying techniques like pruning and quantization to make it more suitable for edge devices. And also extending this work for classification of other cancer types.

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